

Darwin Del Castillo, MD, MPH

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EDUCATION

Master of Public Health (MPH) – Epidemiology
UNIVERSITY OF WASHINGTON

Sep 2023 – Jun 2025
SEATTLE – USA

Doctor of Medicine (MD)
UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS

Mar 2014 – Dec 2020
LIMA – PERU

ADDITIONAL EDUCATION

Professional Certificate in Clinical Research
UNIVERSIDAD PERUANA CAYETANO HEREDIA

Sep 2021 – Apr 2022
LIMA – PERU

SKILLS

Programming: R, Python, SQL (MySQL), SAS

Cloud Computing: Slurm HPC, Linux (bash)

Tools: Quarto, Shiny (R), REDCap, Git (GitHub, GitLab)

Data Standards: ICH E6, HIPAA, GDPR/UK GDPR, ICD-10/ICD-9, SNOMED CT

Packages: tidyverse, survival, lme4, rstan, mice, cmprsk, ivreg, metafor

Statistical methods: Bayesian hierarchical modelling (MCMC), GLM/GLMM, time-to-event models, marginal structural models, propensity score methods, competing risks, multiple imputation, meta-analysis, uncertainty estimation

Languages: English (fluent), Spanish (native)

Core Competencies: Real-world evidence generation, observational study design, scientific writing, stakeholder communication

PROFESSIONAL EXPERIENCE

Research Assistant – Epidemiology
SCHOOL OF HEALTH AND WELLBEING – UNIVERSITY OF GLASGOW

Dec 2025 – Current
GLASGOW – UK

- Estimating causal effects of social policy on health outcomes using survey data with g-methods.
- Modelling the direct and indirect effects of economic determinants of health on all-cause mortality using linked administrative data and doubly robust inference.
- Simulating policy scenarios using a dynamic stochastic model, delivering dashboards to be shared with key stakeholders.

Research Assistant – Clinical Informatics
INSTITUTE FOR HEALTH METRICS AND EVALUATION

Jul 2024 – Jun 2025
SEATTLE – USA

- Developed inputs for a Bayesian model to estimate the burden and clinical outcomes of lower respiratory tract infections worldwide.
- Analysed large-scale RWD, including structured EHR data from 2 countries with 60+ million records, to generate epidemiological summaries of risk factors by sex.
- Developed MCMC-based pipelines with uncertainty propagation to estimate overlap among vulnerable population subgroups in 22 high-income countries to compute summary epidemiological measures.

- Generated real-world evidence from harmonised claims (>5 million encounters) using ICD-10 phenotyping and probabilistic linkage.
- Built regression and survival analysis pipelines in R with quality control for reimbursement policy analyses and healthcare decision-making
- Developed analytical pipelines to evaluate country-level reimbursement policy options

- Analysed longitudinal data from population cohorts and randomised trials in cardiometabolic epidemiology.
- Conducted analyses with biological biomarkers to support research grant proposals and scientific outputs.
- Co-authored 2 scientific manuscripts involving longitudinal analyses and machine learning models.

LICENSURE AND CERTIFICATION

National Registry of Investigators, RENACYT-CONCYTEC, Investigator Level VII (P0160977)	2024
CITI Program, Good Clinical Practice (ICH E6), Credential ID 65079238	2023
CITI Program, Human Subjects Learners, Credential ID 65079240	2023
CITI Program, Biomedical Responsible Conduct of Research, Credential ID 65079239	2023
Peruvian Medical Board, License No. 94534	2021

PUBLICATIONS **ORCID: 0000-0002-8609-0312**

Peer-Reviewed Publications

1. **Del Castillo-Fernández D**, Brañez-Condorena A, Villacorta-Landeo P, Saavedra-Garcia L, Bernabé-Ortiz A, Miranda J. Advances in the investigation of chronic non-communicable diseases in Peru. *An Fac med.* 2021;81(4) doi:10.15381/anales.v81i4.18798.
2. Lapidaire W, Proaño A, Blumenberg C, Loret de Mola C, Delgado CA, **Del Castillo D**, et al. Effect of preterm birth on growth and blood pressure in adulthood in the Pelotas 1993 cohort. *Int. J. Epidemiol.* 2023;52(6):1870–1877 doi:10.1093/ije/dyad084.
3. Vidyasagaran AL, Ayesha R, Boehnke JR, Kirkham J, Rose L, Hurst JR, et al. Core outcome sets for trials of interventions to prevent and to treat multimorbidity in adults in low and middle-income countries: the {COSMOS} study. *BMJ Glob. Health.* 2024;9(8):e015120 doi:10.1136/bmjgh-2024-015120.
4. Bazo-Alvarez JC, **Del Castillo D**, Piza L, Bernabé-Ortiz A, Carrillo-Larco RM, Smeeth L, et al. Multimorbidity patterns, sociodemographic characteristics, and mortality: Data science insights from low-resource settings. *Am. J. Epidemiol.* 2026;194(4):901–911 doi:10.1093/aje/kwae466.